

Immunoglobulin A Nephropathy and Immunoglobulin A Vasculitis (Henoch-Schönlein Purpura)

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IgA NEPHROPATHY

Definition

Immunoglobulin A (IgA) nephropathy (IgAN) is a mesangial proliferative glomerulonephritis (GN) characterized by diffuse mesangial deposition of IgA. IgAN was first recognized in 1968 by Jean Berger when immunofluorescence (IF) techniques were introduced for the study of kidney biopsy specimens. IgAN is unique among glomerular diseases in being defined by the presence of an immune reactant rather than by any other morphologic feature on kidney biopsy, and the light microscopy changes are variable. IgAN is the most prevalent pattern of glomerular disease seen in most Western and Asian countries where kidney biopsy is widely practiced. It is likely that IgAN is not a single entity but rather a common response to various injurious mechanisms.

Etiology and Pathogenesis

Although there has been much recent progress in understanding the pathogenesis of IgAN, it is not certain that all subjects with IgAN share a single common process leading to mesangial IgA deposition. [Fig. 24.1](#) summarizes some key elements involved in the pathogenesis of IgAN.¹

IgA Immune System

IgA is the most abundant immunoglobulin in the body and is chiefly concerned with mucosal defense.² It has two subclasses, IgA1 and IgA2. Mucosal antigen challenge provokes *polymeric* IgA (pIgA) production by plasma cells of the mucosa-associated lymphoid tissue (MALT); the pIgA is then transported across epithelium into mucosal fluids, where it is released after coupling to secretory component as *secretory* IgA (sIgA). The function of circulating IgA is less clear; it is derived from bone marrow and mostly is *monomeric* IgA1 (mIgA1). Circulating IgA1 is cleared by the liver through hepatocyte asialoglycoprotein receptors and Kupffer cell Fcα receptors.

The regular recurrence of IgAN after kidney transplantation in patients with prior underlying IgAN implies an abnormality in the host IgA immune system (see [Chapter 113](#)). Serum IgA levels, both mIgA and pIgA, are increased in only one-third of patients with IgAN and not useful for diagnosing IgAN. However, high serum IgA itself is not sufficient to cause IgAN. High circulating levels of monoclonal IgA in myeloma or polyclonal IgA in acquired immunodeficiency syndrome (AIDS) only infrequently provoke mesangial IgA deposition and thus in IgAN, changes to the physicochemical properties of IgA, rather than the absolute amount in the circulation, probably cause mesangial deposition and glomerular injury.

IgA Glycosylation

A striking feature of IgAN is altered glycosylation of the hinge region of IgA1. IgA1 carries distinctive O-linked sugars at its hinge region; IgA2

has no hinge and carries no such sugars. There is good evidence that circulating IgA1 in IgAN is enriched for IgA1 O-glycoforms with reduced galactosylation of the IgA1 hinge region; these glycoforms are collectively named galactose-deficient IgA1 (gd-IgA1).¹ O-galactosylation of IgA1 has a clear genetic basis, but it is unclear whether these same genes are linked to the development of IgAN.³ This enrichment of gd-IgA1 in the serum is because of an excess of mucosal IgA1 entering the systemic circulation, either directly from the MALT or after translocation of mucosal lymphocytes to the bone marrow. The latter is consistent with experiments in which immortalized lymphocytes from patients with IgAN continued to produce dimeric and polymeric IgA with altered galactosylation.⁴

Mesangial IgA1 in IgAN also has increased gd-IgA1.^{5,6} IgA and IgG antibodies with specificity for gd-IgA1 have been identified in IgAN, promoting formation of circulating immune complexes.⁷ Mesangial IgA1 may result from deposition of these circulating IgA1 immune complexes or of altered IgA1 interactions with matrix proteins and mesangial cell or monocyte Fc receptors. There also may be impaired clearance of gd-IgA1 through inhibition of its interactions with hepatic and circulating myeloid cell IgA receptors.

Increased serum gd-IgA1 is also found in the unaffected relatives of subjects with IgAN, suggesting that gd-IgA1 is necessary but not sufficient for the development of IgAN, which requires a “second hit,” such as an event that provokes induction of autoimmunity or an encounter with microbial pathogens that triggers a humoral immune response, resulting in IgA cross-reactive antibodies.

Origin of Mesangial IgA

Evidence supporting the importance of mucosal-derived IgA as a major source of mesangial IgA in IgAN includes the close association between mucosal inflammation and episodes of nephritis. In addition, mesangial IgA in IgAN is predominantly pIgA1. IgAN patients exhibit elevated serum levels of sIgA and levels of IgA antibodies specific for food antigens, mucosal vaccines, and gut-associated bacteria. Mucosal hyperresponsiveness to a variety of food antigens has been noted in patients with IgAN.^{1,2} Furthermore, the composition of the gut microbiome appears different in IgAN, and gut bacteria are capable of modulating IgA class switching, the amount of mucosal IgA production, and IgA1 O-galactosylation in the MALT. Genome-wide association studies have identified a number of risk alleles that are either directly associated with risk of inflammatory bowel disease, maintenance of the intestinal epithelial barrier, or response to mucosal pathogens and which converge on the intestinal immune network for IgA production.^{8,9}

Role of Infection

The clinical association of visible hematuria with upper respiratory tract infection in IgAN indicates the mucosa may be a site of entry

Pathogenesis of IgA Nephropathy

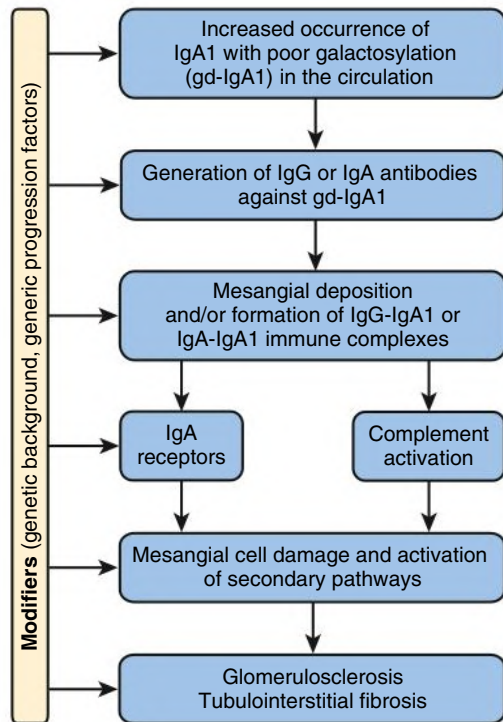


Fig. 24.1 Pathogenesis of Immunoglobulin A (IgA) Nephropathy. Proposed mechanisms leading to mesangial deposition of abnormally glycosylated IgA1 and mesangial injury. (Modified from Floege J. The pathogenesis of IgA nephropathy: what is new and how does it change therapeutic approaches? *Am J Kidney Dis.* 2011;58:992–1004.)

for foreign antigens or alternatively may be a site for nonspecific activation of the innate immune system that enhances kidney injury. There have been occasional reports of IgAN in association with infection, both bacterial (e.g., *Campylobacter*, *Yersinia*, *Mycoplasma*, *Haemophilus*) and viral (e.g., cytomegalovirus, adenovirus, coxsackievirus, Epstein-Barr virus). A severe form of IgAN has been reported in association with staphylococcal infection (see [Chapter 57](#)). However, no organism has been consistently implicated in typical cases of IgAN. It is likely that exposure to a variety of microbial pathogens can result in the development of IgA cross-reactive antibodies, which, in subjects with a propensity for high serum gd-IgA1 levels, results in IgA immune complex formation and mesangial IgA deposition.

Glomerular Injury After IgA Deposition

Polymeric IgA deposition in the mesangium is typically followed by mesangial proliferative GN. Co-deposition of IgG and complement are not necessary for development of GN, but when present have been associated with greater glomerular injury. Complement deposits are usually C3 without C1q. Complement activation occurs via both the mannose-binding lectin pathway and alternative pathway.

Once deposited, pIgA binds to and activates mesangial cells, leading to a cascade of molecular events that result in the histopathologic features making up the MEST-C classification (see later).¹⁰ Importantly, each of these features independently associates with progressive disease. In brief, mesangial cell IgA binding triggers the release of proinflammatory and chemotactic mediators, which act

locally in the glomerulus, leading to mesangial cell proliferation (M) and recruitment of inflammatory cells into the glomerulus (E), occasionally resulting in crescent formation (C). These mediators also cross the glomerular basement membrane (GBM) and directly influence podocyte function, altering podocyte gene expression and glomerular permeability, increasing filtration of pIgA, podocyte damage (glomerular cell crosstalk), and segmental glomerulosclerosis (S).¹¹ In addition to effects within the glomerulus, glomerular-derived cytokines, along with filtered pIgA, are capable of activating tubular epithelial cells (glomerulotubular crosstalk), driving tubulointerstitial inflammation and fibrosis (T) via generic biochemical pathways (see [Chapter 81](#)).

Animal Models of IgA Nephropathy

Animal IgA does not have the same characteristics as human IgA1, and some animals also have IgA clearance mechanisms distinct from those in humans. Therefore, animal models, even if they provoke mesangial IgA deposits, are not particularly informative about the mechanisms that underlie human mesangial pIgA1 deposition, although they have provided many insights into events after IgA deposits have developed.

Genetic Basis of IgA Nephropathy

Urine abnormalities increase in frequency among the relatives of patients with IgAN, although IgAN is rarely found in multiple generations of the same pedigree. More than 90% of all cases of IgAN appear to be sporadic.

Large worldwide genome-wide association studies have identified genetic modulators that seem to affect the prevalence of sporadic IgAN and modulate its course.^{8,9} Variations in major histocompatibility gene loci (HLA-DR, -DQ, -DP, and HLA-B) have consistently been identified. Other gene loci are less consistent and include inflammatory mediators (tumor necrosis factor and α -defensin), gene loci affecting complement factor H, innate immunity, IgA-regulating cytokines, and mucosal integrity.^{8,9}

Other Modulators of the Course of IgA Nephropathy

Many cases of IgAN never come to medical attention, and among those in whom the diagnosis is obtained by kidney biopsy, only a minority develop end-stage kidney disease (ESKD). Thus, in addition to genetic factors, there must be potent further modulators of the course of IgAN, such as essential hypertension, obesity, or smoking (see [Fig. 24.1](#); see also [Natural History](#)).

Epidemiology

IgA nephropathy is typically the most prevalent pattern of glomerular disease in countries where kidney biopsy is widely available. The incidence rate has been estimated at 2 to 10 per 100,000 person-years. The prevalence of IgAN is higher in Asian populations (45 cases per million population/year in Japan) than in Whites (31 cases per million population/year in France).^{12,13} The geographic variation has been associated with the presence of particular gene alleles that protect from IgAN.^{8,9} ([Fig. 24.2](#)). This racial predisposition is maintained in other locations; in the United States, IgAN is less common in Blacks than in Whites of European origin. Perceived prevalence of IgAN also may be influenced by attitudes to the investigation of microhematuria. A country with an active program of routine urine testing will inevitably identify more individuals with urine abnormalities, but IgAN will be identified only if kidney biopsy is performed. Even then, the prevalence of IgAN will be underestimated; a study of kidney donors suggests that the prevalence of IgAN with mesangial proliferative changes and glomerular C3 deposits in the general population in Japan may be 1.6%.¹⁴ This suggests that the vast majority of patients with IgAN never come to medical attention and spontaneously remit.

Geographic Variation in Genetic Risk for IgA Nephropathy

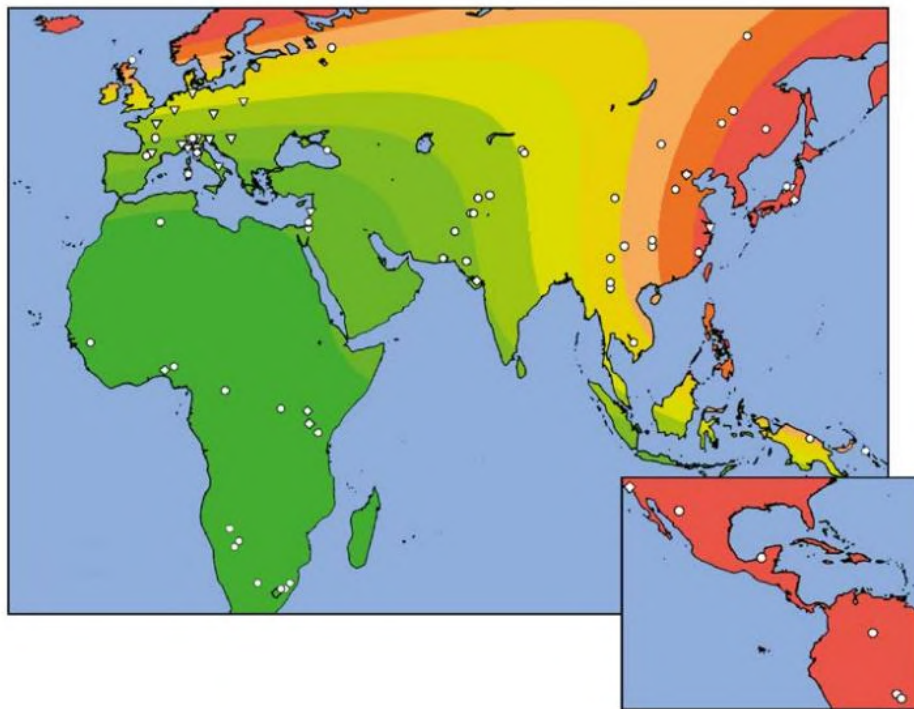


Fig. 24.2 Geographic Variation in Genetic Risk for Immunoglobulin A (IgA) Nephropathy. Genome-wide association studies indicate a geographic gradient of risk from *green* (low risk) to *red* (high risk). (From Kiryluk K, Li Y, Sanna-Cherchi S, et al. Geographic differences in genetic susceptibility to IgA nephropathy: GWAS replication study and geospatial risk analysis. *PLoS Genet.* 2012;8:e1002765.)

Clinical Manifestations

IgA Nephropathy

The wide range of clinical presentations of IgAN varies in frequency with age (Fig. 24.3). No clinical pattern is pathognomonic of IgAN. In populations of White descent, IgAN is more common in males than females by a ratio of 2:1 or 3:1, whereas the ratio approaches 1:1 in most Asian populations.

Macroscopic Hematuria

In 40% to 50% of patients with IgAN, the clinical presentation is episodic macroscopic hematuria, most frequently in the second decade of life. The urine is usually brown rather than red, and clots are unusual. There may be loin pain caused by kidney capsular swelling. Hematuria usually follows intercurrent mucosal infection, typically in the upper respiratory tract (the term *synpharyngitic hematuria* has been used) or occasionally in the gastrointestinal (GI) tract. Hematuria is usually visible within 24 hours of the onset of the symptoms of infection, differentiating it from the 2- to 3-week delay between infection and subsequent hematuria in postinfectious (e.g., poststreptococcal) GN. The macroscopic hematuria resolves spontaneously over a few days. Microscopic hematuria persists between attacks. Most patients have only a few episodes of frank hematuria, which become less frequent and resolve over a few years or sooner. Such episodes may be associated with acute kidney injury (AKI) characterized by tubular injury that is usually reversible.

Asymptomatic Hematuria and Proteinuria

As many as 30% to 40% of cases of IgAN are identified by detection of microhematuria with or without albuminuria (usually <2 g/24 hours)

Clinical Presentations of IgA Nephropathy and IgA Vasculitis in Relation to Age

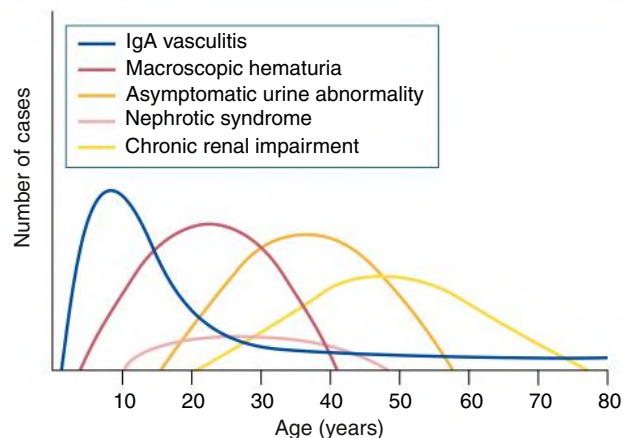


Fig. 24.3 Age in Clinical Presentation of Immunoglobulin A Nephropathy (IgAN) and IgA Vasculitis (IgAV). IgAV is most common in childhood but may occur at any age. Macrohematuria is very uncommon after age 40 years. The importance of asymptomatic urine abnormality as the presentation of IgAN will depend on attitudes to routine urine testing and renal biopsy. It is unclear whether patients presenting late with chronic renal impairment have a disease distinct from that of those presenting younger with macrohematuria. (Data from patients presenting in Leicester, UK, 1980–1995.)

TABLE 24.1 Diseases Reported in Association With IgA Nephropathy

Disease Group	Common	Reported	Rare
Rheumatic and autoimmune disease	Ankylosing spondylitis Rheumatoid arthritis Reiter syndrome Uveitis	Behçet syndrome, ^a Takayasu arteritis, ^b myasthenia gravis	Sicca syndrome
Gastrointestinal disease	Celiac disease	Ulcerative colitis	Crohn disease Whipple disease
Hepatic disease	Alcoholic liver disease Nonalcoholic cirrhosis Schistosomal liver disease		
Lung disease	Sarcoid		Pulmonary hemosiderosis
Skin disease	Dermatitis herpetiformis		
Malignancy		IgA monoclonal gammopathy	Bronchial carcinoma, renal carcinoma, laryngeal carcinoma, mycosis fungoides, Sézary syndrome
Infection	HIV, hepatitis B (in endemic areas)	Brucellosis	Leprosy
Miscellaneous		Wiskott-Aldrich syndrome ^c	

Rare associations have been made in one or two reported cases only. In a disease as common as IgA nephropathy, it is therefore uncertain whether these are truly related.

^aBehçet syndrome: systemic vasculitis typified by orogenital ulceration and chronic uveitis.

^bTakayasu arteritis: systemic vasculitis involving the aorta and its major branches, most often found in young women.

^cWiskott-Aldrich syndrome: X-linked disorder in which increased serum IgA is associated with the triad of recurrent pyogenic infection, eczema, and thrombocytopenia.

HIV, Human immunodeficiency virus; IgA, immunoglobulin A.

during routine urine testing among asymptomatic individuals. The number of patients identified in this way will depend on local attitudes to urine testing and on the use of kidney biopsy in patients with isolated microscopic hematuria. Most patients with IgAN are asymptomatic and would be identified only when the urine is tested. If such patients with very early or mild IgAN are identified by kidney biopsy, their 20-year prognosis for kidney survival is very good, but 5% will progress to chronic kidney disease (CKD) stages IV to V,¹⁵ stressing the need for very long-term follow-up.

Proteinuria and Nephrotic Syndrome

It is rare for proteinuria to occur without microscopic hematuria. Although nephrotic-range proteinuria may occur, in particular in the presence of uncontrolled hypertension or marked obesity, full-blown nephrotic syndrome is uncommon, occurring in only 5% of all patients with IgAN. Nephrotic syndrome may occur early in the course of the disease, with minimal glomerular change (in such cases, IgAN may coexist with minimal change disease [MCD]) or with active mesangial proliferative GN. Alternatively, it may occur as a late manifestation of advanced chronic glomerular scarring.

Acute Kidney Injury

Although AKI is uncommon in IgAN (<5% of all cases), one study reports that it may be the manifestation in up to 40% of patients older than 65 years.¹⁶ The proportion of patients who progressed to ESKD was significantly higher in those presenting with AKI than in patients presenting without AKI. AKI develops by three distinct mechanisms. There may be acute severe immune and inflammatory injury with necrotizing GN and crescent formation (crescentic IgAN); this may be the first presentation of IgAN, or it may be superimposed on established, less aggressive disease. Alternatively, AKI can occur with mild glomerular injury when heavy glomerular hematuria leads to tubule occlusion by red blood cells (RBCs). Third, especially in elderly patients, chronic IgAN will predispose to AKI from a variety of incidental kidney insults, including drug-induced acute tubulointerstitial nephritis (see [Chapter 64](#)).

Chronic Kidney Disease

Some patients already have kidney impairment and hypertension when they are first diagnosed with IgAN. These patients tend to be older, and they probably have longstanding disease that previously remained undiagnosed because the patient did not have frank hematuria or undergo routine urinalysis. Hypertension is common, as in other chronic glomerular disease; accelerated hypertension occurs in 5% of patients.

Clinical Associations With IgA Nephropathy

Mesangial IgA deposition, which does not necessarily equal IgAN, is a frequent finding in autopsy studies in chronic liver disease. Although particularly associated with alcoholic cirrhosis, mesangial IgA deposition can occur in other chronic liver diseases, including those caused by hepatitis B and schistosomiasis. It is thought to result from impaired clearance of IgA by the Kupffer cells, which express Fcα receptors, and hepatocytes, which express the asialoglycoprotein receptor. Secondary IgAN is described as a form of IgAN associated with other diseases (e.g., liver cirrhosis, human immunodeficiency virus [HIV], and inflammatory bowel disease). There are no established definitions or different histologic findings in secondary IgAN cases, so the basis for the term is questionable.¹⁷ Clinical evidence of kidney disease is more common than previously appreciated, but patients rarely develop ESKD.

A number of case reports have associated IgAN with HIV infection and AIDS. It is not clear whether the polyclonal increase in serum IgA, which is a feature of AIDS, is the predisposing factor.

There are case reports associating IgAN with many other conditions, particularly with a number of immune and inflammatory diseases ([Table 24.1](#)). Their relationship to abnormalities of the IgA immune system is not always clear, and some may represent the coincidental development of unrelated but relatively common conditions. In a Swedish population-based cohort study, it was observed that patients with IgAN had an increased risk of inflammatory bowel disease both before and after their nephropathy diagnosis; the presence of inflammatory bowel disease increased the risk of progression to ESKD.¹⁸

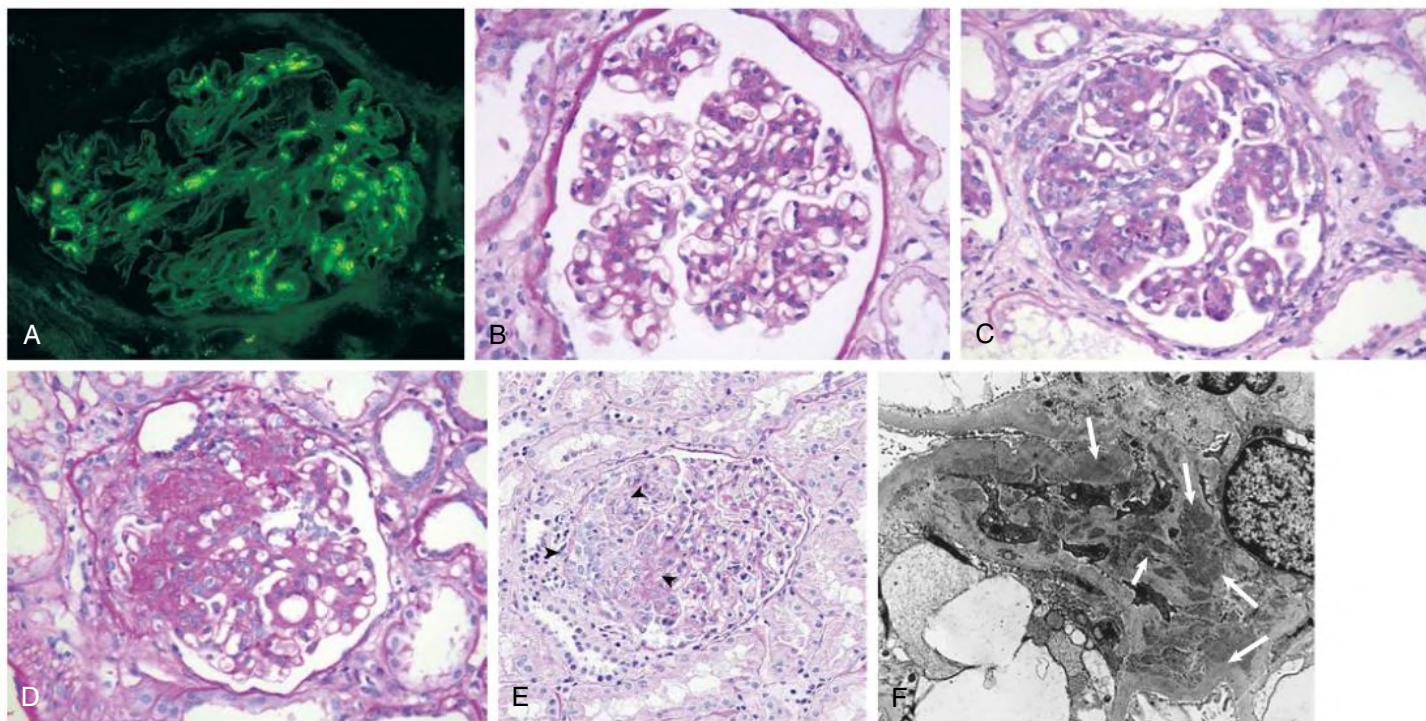


Fig. 24.4 Renal Pathology in IgA Nephropathy (IgAN). (A) Diffuse mesangial IgAN seen on indirect immunofluorescence with fluorescein isothiocyanate–anti-IgA (magnification $\times 3300$). (B) Diffuse mesangial hypercellularity (M1, Oxford classification). (C) Endocapillary hypercellularity (E1). (D) Segmental sclerosis (S1). (E) Crescentic lesion. (B–E, Light microscopy with PAS reaction, $\times 3300$.) (F) Mesangial electron-dense deposits (arrows) (electron micrograph, $\times 316,000$). (B–D, Courtesy Professor I. Roberts. E, Courtesy Professor P. Boor.)

Pathology

Immune Deposits

Diffuse mesangial IgA is the defining hallmark of IgAN (Fig. 24.4A). C3 is co-deposited in up to 90% of cases. IgG in 40% and IgM in 40% of cases also may be found in the same distribution. IgA also may deposit along capillary loops, a pattern more common in IgA vasculitis (IgAV); in IgAN, this pattern is associated with a worse prognosis. Mesangial deposits of terminal pathway complement complex C5b-9 are commonly observed in IgAN. Evidence of classical pathway activation, such as C1q deposits, is usually lacking in kidney biopsy specimens of patients with IgAN. Properdin co-deposits with IgA and C3 in 75% to 100% of patients and factor H in 30% to 90% of patients, indicating alternative complement pathway activation. MBL co-deposits with IgA in 17% to 25% of IgAN biopsies, and glomerular and arteriolar C4d as an indicator of the lectin pathway activation has been associated with more severe kidney damage.¹⁹ Disappearance of IgA deposits after prolonged clinical remission has been documented in both children and adults. About one-third of the patients also have deposits of sIgA in the mesangium and are characterized by more severe disease.²⁰

Light Microscopy

Light microscopy changes are remarkably variable and do not correlate topographically with the IgA deposits. There can be almost normal glomerular architecture, mesangial hypercellularity that may be diffuse (see Fig. 24.4B) or segmental, or, in rare cases, focal segmental necrotizing GN with extracapillary proliferation. Typical cases are characterized by an increase in mesangial cells and mesangial matrix with normal-appearing capillary loops, although endocapillary hypercellularity can occur (see Fig. 24.4C). Focal segmental or global glomerular sclerosis indicates that the disease has been ongoing for some time (see Fig. 24.4D). In addition to glomerular changes, the preglomerular arterial vessels often exhibit wall hyalinosis and subintimal fibrosis even in

patients with only mild arterial hypertension. In longstanding disease, tubulointerstitial inflammation leads to interstitial fibrosis and tubular atrophy in a pattern no different from that of other progressive glomerular diseases. On occasion, IgAN and minimal change nephrotic syndrome coincide (see Differential Diagnosis), in which case light microscopy is normal but there are mesangial IgA deposits.

Morphology helps to predict prognosis in patients with slowly progressive disease. The Oxford (MEST-C) classification of IgAN is now widely accepted.¹⁰ This classification identifies five features of prognostic value and can be easily scored on light microscopy: mesangial hypercellularity (M1 when present, M0 when absent), endocapillary hypercellularity (E1), segmental sclerosis (S1), three degrees of tubular atrophy and interstitial fibrosis (T0, T1, T2), and three degrees of cellular and/or fibrocellular crescents (C0, C1, C2). Examples of these glomerular features are shown in Fig. 24.4.

Histologic patterns of injury in AKI include tubular occlusion by RBCs with acute tubular epithelial injury in AKI associated with macroscopic hematuria (Fig. 24.5) or necrotizing GN and cellular crescent formation. Such crescentic IgAN may develop on a histologic background of established chronic kidney injury caused by IgAN or may be the first presentation of IgAN. Small numbers of crescents may be seen in hypertensive patients with stable kidney function and no other pathologic evidence of severe glomerular inflammation; the term *crescentic IgAN* should not be used for such patients, in whom the prognosis is often favorable.

Electron Microscopy

Electron-dense deposits correspond to the mesangial (or capillary loop) IgA (see Fig. 24.4E). Typically, electron-dense deposits are confined to mesangial and paramesangial areas, although subepithelial and subendothelial deposits also can be seen. Up to one-third of patients will have some focal thinning of the GBM. On occasion, there

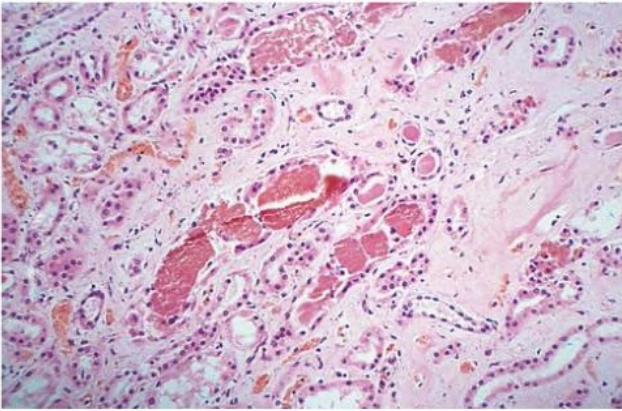


Fig. 24.5 Acute Kidney Injury in Immunoglobulin A Nephropathy. Tubular occlusion by red blood cells. This appearance may be associated with only minor glomerular changes. (H&E, $\times 300$.)

will be extensive GBM thinning, suggesting a coincident diagnosis of thin basement membrane nephropathy (see [Chapter 48](#)).

Differential Diagnosis

The diagnosis of IgAN or IgAV nephritis requires identification of mesangial IgA in the glomeruli. Therefore, it cannot be made without a kidney biopsy, no matter how suggestive the clinical presentation. Serum IgA is often increased, and there may be IgA in cutaneous blood vessels in IgAN and in both affected and unaffected skin in IgAV. Neither finding, however, is reliable enough to support the diagnosis without a kidney biopsy. Serum complement components are normal in most patients.

Mesangial IgA occurs in other conditions and can usually be differentiated by clinical, serologic, and histologic criteria ([Box 24.1](#)). The light microscopic features alone are not diagnostic of IgAN.

An important differential diagnosis is IgA-dominant acute postinfectious GN (APIGN), in which IgA is the dominant immunoglobulin in glomerular deposits (see [Chapter 57](#)). It usually occurs in association with staphylococcal infections. Diabetes is a key risk factor. Compared with IgAN, patients with IgA-dominant APIGN are more likely to be older and have AKI, documented staphylococcal infection, hypocomplementemia, diffuse glomerular endocapillary hypercellularity with prominent neutrophil infiltration on light microscopy, stronger IF staining for C3 than IgA, and the presence of subepithelial humps on electron microscopy (EM).²¹ This entity has a poor prognosis.

Hematuria

Nonglomerular causes of hematuria, particularly stones and neoplasia, must be excluded by appropriate investigations (see [Chapter 63](#)). In its most characteristic clinical setting—recurrent macroscopic hematuria coinciding with mucosal infection in a man in the second or third decade of life—the diagnosis can be strongly suspected. Such a diagnosis, however, cannot be made without a biopsy because recurrent macroscopic hematuria also occurs in other glomerular diseases (e.g., Alport syndrome and membranoproliferative GN), particularly in children and young adults. In young adults, thin basement membrane nephropathy is the most important differential diagnosis for isolated microhematuria.

Nephrotic Syndrome

Patients with IgAN occasionally develop nephrotic syndrome, which is indistinguishable from that in MCD. There is a sudden onset of nephrosis, with biopsy evidence of glomerular epithelial cell foot process effacement and a prompt complete remission of proteinuria in response to corticosteroids. Only hematuria and mesangial IgA

BOX 24.1 Differential Diagnosis of Immunoglobulin A (IgA) Nephropathy: Conditions Associated With Mesangial IgA Deposition

- IgA nephropathy
- IgA vasculitis
- Lupus nephritis^a
- Alcoholic liver disease
- IgA monoclonal gammopathy
- Schistosomal nephropathy
- IgA-dominant *Staphylococcus*-associated glomerulonephritis (more common in diabetics)

^aDistinguishing lupus nephritis (especially International Society of Nephrology/Renal Pathology Society classes II and III) may cause difficulty. The finding of C1q deposition is useful. It indicates classical pathway involvement found in lupus nephritis but not in IgAN.

deposits persist after treatment. This pattern occurs particularly in children. These patients are usually regarded as having two separate common glomerular diseases: IgAN and MCD.

Other patients with IgAN may develop nephrotic syndrome with more structural glomerular damage and lack the response to corticosteroids. The clinical differential diagnosis includes common causes of nephrotic syndrome appropriate for the age of the patient (see [Chapter 16](#)).

Chronic Kidney Disease: Hypertension, Proteinuria, Kidney Impairment

In this context, IgAN is clinically indistinguishable from many forms of CKD. The kidney biopsy may be diagnostic by identifying mesangial IgA, even when structural damage is so advanced on light microscopy that it has the nonspecific features of ESKD.

Acute Kidney Injury

When AKI occurs in a patient known to have IgAN, kidney biopsy should be performed unless there is rapid improvement in kidney function, after at least 5 days from the onset of kidney function worsening, in response to supportive care and vigorous hydration. Kidney biopsy may be required to differentiate the tubular occlusion and acute tubular necrosis that occasionally follow heavy glomerular hematuria from crescentic IgAN or other coincidental causes of AKI (see [Fig. 24.5](#)).

Natural History

IgA Nephropathy

The overall prognosis of IgAN differs based on the clinical presentation. If diagnosed very early, only about 5% will develop CKD stages 4 to 5 after 20 years.¹⁵ In the more typical clinical presentation, where many patients already exhibit some degree of GFR decline, at 20 years later one-fourth of patients will have ESKD and a further 20% will have progressive impairment of kidney function.^{22,23} A Swedish study suggested that patients with IgAN died 6 years earlier than people without the disease; most deaths were from cardiovascular causes.²⁴

Although active investigation of microhematuria will increase the size of the cohort of patients found to have IgAN, it will include more with a good prognosis,¹⁵ thus altering the perceived risk for disease progression. Episodes of macrohematuria do not confer a worse prognosis. This may indicate that such episodes occur only early in the natural history of the disease and patients doing less well from the point of diagnosis in fact were identified at a later stage in their disease. This is also suggested by the adverse prognosis associated with older age at diagnosis.

The risk for ESKD is not uniform. As in any chronic glomerular disease, hypertension, proteinuria, and reduced glomerular filtration rate

BOX 24.2 Prognostic Markers Used to Calculate the Risk of Progressive GFR Loss in Immunoglobulin A Nephropathy

Estimated GFR at biopsy (mL/min/1.73 m²)
 Systolic blood pressure at biopsy (mm Hg)
 Diastolic blood pressure at biopsy (mm Hg)
 Proteinuria at biopsy (g/day)
 Age at biopsy (years)

Race

White
 Chinese
 Japanese
 Other

Use of ACE Inhibitor or ARB at Time of Biopsy

No
 Yes

MEST M-Score

0
 1

MEST E-Score

0
 1

MEST S-Score

0
 1

MEST T-Score

0
 1
 2

Immunosuppression Use at or Before Biopsy

No
 Yes

Using clinical and histologic data at biopsy users can determine a 50% decline in eGFR or kidney failure at selected time intervals. The tool is not validated for use with data obtained remotely from the time of biopsy.

ACE, Angiotensin-converting enzyme; ARB, angiotensin receptor blocker; GFR, glomerular filtration rate; MEST, mesangial (M) and endocapillary (E) hypercellularity, segmental sclerosis (S), and interstitial fibrosis/tubular atrophy (T).

Modified from Barbour SJ, Coppo R, Zhang H, et al. Evaluating a new international risk-prediction tool in IgA nephropathy. *JAMA Intern Med.* 2019;179:942–952.

(GFR) at presentation, as well as histologic evidence of glomerular and interstitial fibrosis, are all associated with worse prognosis (Box 24.2). Hyperuricemia, smoking, and increased body mass index are also independent risk factors for progression. However, during follow-up, only hypertension and proteinuria, in particular time-averaged proteinuria, are reliable predictors of risk for progression. Risk for progression is low when proteinuria remains below approximately 0.2 g per 24 hours with normal blood pressure.^{15,22,23,25,26} Another risk predictor for kidney disease progression is persistent hematuria, whereas hematuria remission was associated with improved kidney outcomes in IgAN among patients with persistent proteinuria.²⁷

If a kidney biopsy documents IgAN in patients with mild disease (i.e., those manifesting with isolated microhematuria, little or no proteinuria, normotension, and normal GFR), the 7- to 10-year prognosis is mostly good.^{15,25,26} However, up to 40% of patients will develop increasing proteinuria and up to 5% will lose GFR over this period, implying the need for regular follow-up in such patients. Large studies with long-term follow-up indicate a slow attrition.

Pathologic findings and clinical findings together inform prognosis.²⁸ The MEST-C classification of IgAN identified mesangial hypercellularity, endocapillary proliferation, segmental sclerosis, tubular atrophy and interstitial fibrosis, as well as crescents, as adding prognostic information even when adverse clinical features (proteinuria, hypertension, GFR) are known at presentation and during follow-up.¹⁰

The International IgAN Prediction Tool (available at www.qxmd.com) includes two validated Cox survival models that predict a 50% decline in estimated GFR (eGFR) or ESKD using clinical risk factors and Oxford MEST histology scores.²⁹

Transplantation

Recurrent IgA Nephropathy

Patients with IgA nephropathy are ideal candidates for kidney transplant because they are often relatively young and with few comorbidities. Transplant registry data show that transplant outcome is not affected for the first 10 years if IgAN is the patient's primary kidney disease; thereafter, however, recurrent disease may lead to accelerated graft loss (see Chapter 113). Mesangial IgA deposits recur in the donor transplant kidney in up to 60% of patients with IgAN.³⁰ They may occur within days or weeks, but the risk increases with the duration of the transplant. The deposits seem benign in the short term and are not often associated initially with light microscopy changes. In pooled series, recurrent IgAN is 30% in living related transplants versus 23% in cadaveric grafts, but this does affect graft survival only beyond 10 years after transplantation.³⁰ Living related donation should not be discouraged. However, any urinary abnormality in a potential related donor requires thorough evaluation, including kidney biopsy if necessary. Recurrence of crescentic IgAN with rapid graft failure occurs infrequently and is generally resistant to treatment.

In a few unwitting experiments, cadaver kidneys with IgA deposits have been transplanted into recipients without IgAN. In all cases, the IgA rapidly disappeared, supporting the concept that abnormalities in IgAN lie in the IgA immune system and not in the kidney.

Treatment

Although specific early treatment intervention might influence the IgA immune system abnormalities that underlie IgAN, the mechanisms of chronic disease progression are unlikely to be unique. Therefore, studies of such patients with IgAN may provide information applicable to many forms of chronic GN for which IgAN is the paradigm.

The balance of risk to benefit for immunosuppressive therapy is often unfavorable in patients with IgAN, except in the unusual circumstance of crescentic IgAN.

Fig. 24.6 provides a treatment algorithm for IgAN that is based on the current Kidney Disease: Improving Global Outcomes (KDIGO) guidelines.^{30a} The approach distinguishes among the following³¹:

- The “low-risk” patient, with minor urinary abnormalities, normal GFR, and normotension, who needs only sporadic follow-up for a prolonged period (>10 years)
- The “intermediate-risk” patient, with significant proteinuria, hypertension, and slow reduction of GFR, who particularly benefits from comprehensive supportive care

Treatment Recommendations for IgA Nephropathy

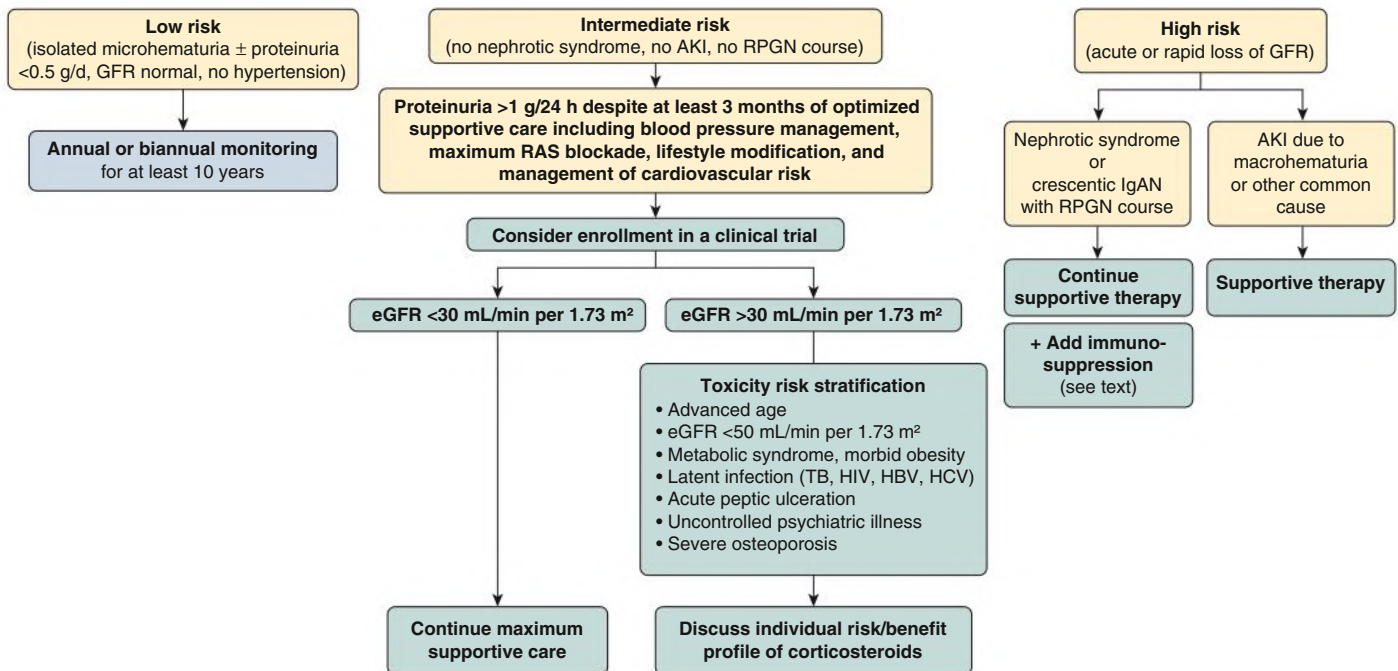


Fig. 24.6 Treatment Recommendations for Immunoglobulin A Nephropathy (IgAN). AKI, Acute kidney injury; GFR, glomerular filtration rate; HBV, hepatitis B virus; HCV, hepatitis C virus; HIV, human immunodeficiency virus; RPGN, rapidly progressive glomerulonephritis; TB, tuberculosis. (Modified from Floege J, Feehally J. Treatment of IgA nephropathy and Henoch-Schonlein nephritis. *Nat Rev Nephrol*. 2013;9:320–327.)

- The “high-risk” patient, with rapid loss of GFR, who may require more aggressive immunosuppression

Slowly Progressive IgA Nephropathy (Intermediate Prognosis)

Minimal evidence exists to suggest the events of progressive glomerular injury are unique to IgAN. Comprehensive supportive therapy following the strategies outlined in [Chapter 82](#) therefore remains central in the therapeutic approach to patients at risk for progressive IgAN (see [Fig. 24.6](#)).

Antihypertensives and proteinuria-lowering drugs. There is compelling evidence for the benefit of lowering blood pressure (BP) in the treatment of chronic progressive glomerular disease such as IgAN. In patients with IgAN, there is also evidence that casual clinic BP readings underestimate BP load, as judged by ambulatory monitoring and echocardiographic evidence of increased left ventricular mass.³² Two controlled trials strongly support the use of angiotensin-converting enzyme (ACE) inhibitors in patients with IgAN as first-choice hypotensive agents to lower BP.^{33,34} In a randomized study of IgAN, achieving a mean BP of 129/70 mm Hg prevented the decrease in kidney function over 3 years seen in patients achieving mean BPs of 136/76 mm Hg.³⁵

Observational studies indicate that the risk for progression in IgAN decreases significantly if proteinuria can be reduced to less than 1 g/day by any therapeutic maneuver.²³ Another compelling indication for the role of lowering proteinuria in IgAN is based on the superior effectiveness of ACE inhibitors over other antihypertensive agents.^{33,34} There is no good evidence to recommend dual renin-angiotensin system (RAS) blockade with ACE inhibitors plus angiotensin receptor blockers over monotherapy with either agent alone.³⁶

Fish oil. The favorable effects of omega-3 fatty acids in the form of fish oil include reductions in eicosanoid and cytokine production, changes in membrane fluidity and rheology, reduced platelet aggregability, and reduced proliferation of kidney cells in response to platelet-derived growth factors (PDGF). The evidence for a benefit of fish oil in IgAN is weak; the 2021 KDIGO guidelines no longer support its use.^{30a} However, fish oil is safe except for a decrease in blood coagulability, which is not usually a practical problem, and an unpleasant taste, with flatulence, which may make compliance difficult. Some fish oil preparations contain significant amounts of cholesterol.

Dipyridamole and warfarin. Two randomized controlled trials (RCTs) with dipyridamole and warfarin showed mutually inconsistent results. There was no benefit in one and preserved kidney function in the other. Neither drug is currently recommended in IgAN patients.^{30a}

Newer nonimmunosuppressant approaches. A major reduction in renal endpoints (40% GFR loss, dialysis, death) was observed in IgAN patients treated with dapagliflozin.^{30b} It is likely that inhibition of sodium glucose transporter-2 (SGLT-2) will become standard of care after RAS-blockade has been optimized in high-risk patients with IgAN. Empagliflozin is currently being evaluated in IgAN as well (EMPA-Kidney trial). A Chinese RCT suggested that hydrochloroquine (100–400 mg/day depending on GFR) may be useful to reduce albuminuria, potentially related to the mild antiinflammatory actions of the drug.³⁷ Various other novel treatment approaches, including dual angiotensin plus endothelin receptor blockade (PROTECT trial), endothelin A receptor blockade on top of RAS blockade (ALIGN trial), and various approaches to complement inhibition (e.g., the narsoplimab ARTEMIS or iptacopan APPLAUSE trials) are currently in phase II to III testing.

Immunosuppressive Regimens

Corticosteroids. A meta-analysis and retrospective data suggest that corticosteroids are beneficial in IgAN, in particular in adults with proteinuria exceeding 3 g/day.^{38,39} However, several trials have failed to demonstrate benefit from corticosteroids, and dosage was somewhat lower in two of the three negative trials versus the positive trials.³¹ Several trials did not optimize supportive therapy, in particular RAS blockers, or required that such drugs be temporarily halted before the trial.³¹

The role of corticosteroid therapy has been investigated in two large prospective trials of oral corticosteroids in IgAN added to optimized supportive care (STOP-IgAN and TESTING). In the German STOP-IgAN trial, an extensive 6-month optimization of supportive measures abolished any subsequent short- or long-term benefit from high-dose corticosteroids in adult patients with IgAN with an average baseline GFR of 60 mL/min/1.73 m² and baseline proteinuria around 1.7 g/d.^{40,41} The largely Southeast Asia-based TESTING trial on high-dose corticosteroids in adults with a baseline GFR of 58 mL/min and a baseline proteinuria of approximately 2 g/day was terminated early after an excess of severe, sometimes fatal infections was noted in the corticosteroid arm.⁴² The trial was continued using a 50% lower corticosteroid dosage and demonstrated a considerable yet transient benefit on kidney function, but at the expense of more serious adverse events in the steroid arm.^{42a} These two trials strongly emphasize the value of comprehensively optimizing supportive measures in high-risk patients with IgAN and suggest that corticosteroids may slow loss of renal function, particularly high-risk Asian IgAN patients.

Accordingly, the 2021 KDIGO guidelines suggest that corticosteroids should be initiated in high-risk patients only if proteinuria remains greater than 1 g/day after supportive care has been optimized for 3 to 6 months and only after a thorough discussion of potential adverse events with patients.^{30a} At present, no evidence supports combined intravenous (IV) and oral corticosteroid therapy over a purely oral prednisolone regimen, starting with 1 mg/kg/day for 2 months, then reduced by 0.2 mg/kg/day per month or low dose at 0.4 mg/kg/day methylprednisolone with tapering over 6 months.^{31,42a} KDIGO also points out that corticosteroids should be used very restrictively or be avoided in obese patients (body mass index > 30 kg/m²) or in patients with a GFR of less than 30 to 50 mL/min/1.73 m², diabetes mellitus, latent infection, secondary IgAN, active peptic ulcers, or psychiatric illness.^{30a}

Full-blown nephrotic syndrome may occur when MCD and IgAN coincide, in which case the nephrotic syndrome will be fully and promptly corticosteroid responsive. A trial of high-dose corticosteroid therapy analogous to that used in MCD (see Chapter 18) is therefore justified in patients with IgAN who have nephrotic syndrome associated with minimal glomerular injury.

An alternative approach to corticosteroid dosing has been evaluated. An enteric release corticosteroid (budesonide) preparation administered to patients with intermediate-risk IgAN led to a reduction in proteinuria and better preservation of kidney function in the treated patients.⁴³ Based on the phase III NEFIGARD trial, enteric release budesonide has now obtained a conditional approval for use in high-risk IgAN patients in some countries.

Cyclophosphamide and azathioprine. Cyclophosphamide has been used in combination with warfarin and dipyridamole in two RCTs, with inconsistent results. Both showed modest reduction in proteinuria, but only one preserved kidney function. Cyclophosphamide followed by azathioprine combined with prednisolone preserved kidney function in patients with poor prognosis, although BP control was suboptimal.⁴⁴ The same regimen, however, was ineffective in the STOP-IgAN trial and led to a death

from pulmonary sepsis.⁴⁵ In another recent study, adding azathioprine to corticosteroids in patients with proteinuric IgAN with GFR greater than 50 mL/min/1.73 m² had no added benefit and only increased side effects.⁴⁶ Neither agent is therefore recommended by the 2021 KDIGO guidelines in patients with intermediate-risk IgAN.^{30a}

Other immunosuppressive approaches. Mycophenolate mofetil (MMF) has been used in several RCTs in high-risk patients. Three trials in White patients failed to demonstrate a benefit, whereas studies in Chinese patients noted reduced proteinuria and preserved GFR.^{31,47,48} Whether ethnic differences underlie these discrepant results remains to be clarified. In another Chinese trial, 4 of 32 patients with IgAN receiving MMF plus corticosteroids died of *Pneumocystis pneumonia*.⁴⁹ MMF is not recommended by the 2021 KDIGO guidelines in non-Chinese patients with intermediate-risk IgAN.^{30a}

Cyclosporine has been used in one RCT in IgAN.⁵⁰ Patients showed a reversible decrease in proteinuria along with a decrease in creatinine clearance, suggesting the changes were a hemodynamic effect of cyclosporine rather than an immunomodulating effect.

Rituximab (two infusions of 1 g) failed to reduce levels of undergalactosylated IgA and autoantibodies against this IgA in a small open-label trial in adult patients with high-risk IgAN and did not affect proteinuria over the course of 1 year.⁵¹

Pooled human immunoglobulin has yielded encouraging preliminary results in patients with IgAN who have an aggressive clinical course. Proteinuria was reduced, deterioration of GFR slowed, and histologic activity lessened on repeated kidney biopsies.⁵² No RCT is available for this approach.

Newer immunosuppressive approaches targeting B-cell activation (e.g., B-cell activating factor [BAFF] and/or a proliferation-inducing ligand [APRIL] or using BAFF/APRIL inhibitors such as atacicept) are being studied in patients with IgAN.

Rapidly Progressive IgA Nephropathy (Poor Prognosis)

In this uncommon situation of rapidly progressive kidney failure associated with crescentic IgAN (i.e., not crescentic IgAN associated with stable GFR), the risk-to-benefit ratio most strongly favors intensive immunosuppressive therapy because untreated patients will rapidly progress to ESKD. Treatment has often combined plasma exchange with prednisolone and cyclophosphamide.⁵³ Early clinical response is favorable, as in other crescentic nephritis. Medium-term results, however, are disappointing; kidney survival at 5 years was only 30% and not different in immunosuppressed and nonimmunosuppressed patients.⁵⁴

A subset of patients with circulating IgG-antineutrophil cytoplasmic antibody (ANCA) may have a more favorable response to immunosuppressive therapy similar to that seen in other ANCA-positive crescentic nephritis.⁵⁵ With no RCTs of treatment, it is impossible to be certain which elements of the regimen (corticosteroids, cyclophosphamide, or plasma exchange) are mandatory.

Other Therapeutic Approaches to Progressive IgA Nephropathy

Reduction of IgA production. Tonsillectomy reduces the frequency of episodic hematuria when tonsillitis is the provoking infection. In all other patients with IgAN, tonsillectomy is not routinely recommended^{30a}; there is no role for prophylactic antibiotics. Dietary gluten restriction, used to reduce mucosal antigen challenge, has not been shown to preserve kidney function.²

Prevention and removal of IgA deposits. The ideal treatment of patients with IgAN would remove IgA from the glomerulus and prevent further IgA deposition. This remains a remote prospect while the pathogenesis remains incompletely understood.

Transplant recurrence. There is no evidence that newer immunosuppressive agents have modified the frequency of recurrent

IgA deposits or are of value in recurrent disease. There is, however, registry evidence that transplant outcome is improved if corticosteroids are continued long term and if the immunosuppression includes MMF.^{56,57} In patients with established IgAN recurrence, most clinicians merely optimize supportive care. When crescentic IgAN recurs with rapidly deteriorating graft function, treatment as for primary crescentic IgAN has been used, although evidence of its success is sparse.

IgA VASCULITIS

Definition

IgAV, previously termed Henoch-Schönlein purpura, is a small-vessel vasculitis that predominantly affects children and affects the skin, joints, gut, and kidneys. It is defined by tissue deposition of IgA. Typically, there is clinical involvement in the skin, gut, and kidneys. The nephritis associated with IgAV is characterized by diffuse mesangial IgA deposition. Indeed, the kidney histologic features of IgAV are indistinguishable from those of IgAN.

Epidemiology

In children, IgAV is usually diagnosed on clinical grounds without biopsy confirmation of tissue IgA deposition. Transient urine abnormalities are common in the acute phase. However, only those with persistent urine abnormalities or with more overt kidney disease will come to kidney biopsy. Therefore, the incidence of IgAV is almost certainly underestimated, with many unidentified mild and transient cases. There is no information on geographic variations in IgAV.

Despite some differences in age at onset and natural history of IgAN and IgAV,⁵⁸ there is much evidence to support a close link between the two conditions. Monozygotic twins who developed IgAN and IgAV, respectively, at the same time have been described. The evolution of IgAN into IgAV in the same patient is described in both adults and children, and patients with IgAV and ESKD receiving a kidney transplant may experience recurrent disease in the form of IgAN.

Pathogenesis

Many of the abnormalities of IgA production and handling reported in IgAN are also detected in IgAV, including circulating IgA-rheumatoid factors in 55% of cases, and increased serum gd-IgA1, which is found in IgAV when there is nephritis, but it is not clear whether gd-IgA1 is also predictive of extrarenal manifestations of IgAV. No studies have investigated whether gd-IgA1 is in mesangial deposits in IgAV. There is no animal model for IgAV to facilitate studies of pathogenesis.

Although infection precedes IgAV in up to 50% of cases, no evidence indicates a role for any specific antigen.

Genetics

Subjects with IgAV have been excluded from recent large-scale genome-wide association studies in IgAN, and the genetic background to IgAV has not been systematically studied.

Clinical Manifestations

Although most prevalent in the first decade of life, IgAV may occur at any age. A palpable purpuric rash, which may be recurrent, occurs on extensor surfaces (Fig. 24.7). There may be polyarthralgia (usually without joint swelling) and abdominal pain caused by gut vasculitis. This may be severe, with bloody diarrhea if intussusception develops. In practice, the diagnosis is made by clinical criteria in the great majority of children, in whom IgAV is often a self-limiting illness. In adults, clinical features include purpura, arthritis, and GI symptoms in 95%, 60%, and 50% of patients, respectively.⁵⁹ Kidney involvement in adults with IgAV does not differ from that in isolated IgAN. Tissue



Fig. 24.7 Immunoglobulin A Vasculitis. The rash is a palpable purpuric vasculitis on the lower limbs spreading on extensor surfaces to the buttocks and occasionally to the upper limbs. Histology shows leukocytoclastic vasculitis with immunoglobulin A deposits in blood vessel walls.

BOX 24.3 Treatment Recommendations for Nephritis in IgA Vasculitis

Crescentic nephritis: regimen is as for crescentic IgA nephropathy (see Fig. 24.6).

All other nephritis in IgA vasculitis (including nephrotic syndrome): regimen as for IgA nephropathy (see Fig. 24.6).

Hypertension: angiotensin-converting enzyme inhibitors and angiotensin receptor blockers are agents of first choice; target blood pressure: 130/80 mm Hg if proteinuria is <1 g/24 h, 125/75 mm Hg if proteinuria is >1 g/24 h.

Transplantation: Cadaveric donor may be preferable to living-related donor in children (controversial).

confirmation of IgA deposition by kidney or skin biopsy is necessary to establish the diagnosis of IgAV.

Much kidney involvement in IgAV is transient. Urine abnormalities are noted during the acute presentation but may disappear. Of those referred to a nephrologist, asymptomatic urine abnormality is still the most frequent clinical manifestation. Nephrotic syndrome occurs in 20% to 30% of patients. AKI may develop as a result of crescentic GN.

Pathology

The kidney histopathologic findings in IgAN and IgAV nephritis may be indistinguishable (see Fig. 24.4). The predictive value of the MEST score has not been shown for IgAV with kidney involvement.

Differential Diagnosis

In children, the diagnosis of IgAV is usually made on the basis of clinical criteria; not all patients have a kidney biopsy to confirm tissue IgA deposition. In adults, the differential diagnosis is much wider and includes other forms of systemic vasculitis, requiring diagnosis by clinical, serologic, and histologic characteristics (see Chapter 26).

Natural History

The natural history for IgAV is less clear than for IgAN. Observations are restricted to patients referred for kidney biopsy, which excludes the majority of patients with minor transient kidney involvement, who have an excellent prognosis. The kidney prognosis is worse in adults than in children with IgAV. In adults, up to 40% will have CKD or

ESKD 15 years after biopsy. One series reports an increased mortality from lung and GI malignancy.⁵⁹

Transplantation

IgAV can recur as isolated IgA deposits in the graft (~50% of transplants), as full-blown yet isolated IgAN, or, rarely, as a full recurrence of systemic involvement, including a rash. The clinical and pathologic characteristics of kidney recurrence are similar to those of recurrent IgAN.³⁰ Delay of transplantation after initiation of dialysis does not reduce the risk for IgAV recurrence.

Treatment

Many patients have transient nephritis during the early phase of IgAV, which spontaneously remits and requires no treatment. No RCTs guide the treatment of IgAV. Most therapeutic studies of IgAN exclude those with IgAV, so it is unclear whether the potential treatments for IgAN have a role in IgAV.^{30a} Commonly used treatments for IgAV are shown in [Box 24.3](#).

Rapidly Progressive Chronic Kidney Disease Caused by Crescentic Nephritis

Crescentic nephritis is more common in IgAV than in IgAN, particularly early in the course of the disease. There is little specific information on treatment in adults or children, but regimens based on those for other forms of systemic vasculitis are widely used. These include corticosteroids and cyclophosphamide, with the addition of plasma exchange or pulse methylprednisolone in some cases. However, a French RCT in adults with severe IgAV failed to detect a benefit of cyclophosphamide plus steroids over steroids alone.⁶⁰

Active IgA Vasculitis Without Kidney Failure

There is little information about less aggressive IgAV. Corticosteroids alone, although often considered more effective than in IgAN, have never been shown to be beneficial for prevention or treatment of kidney involvement in IgAV, although are apparently beneficial for extra-renal manifestations.^{30a} Promising findings with combination therapy of corticosteroids, cyclophosphamide, and antiplatelet agents have been reported in only small, nonrandomized studies.^{30a} A nonrandomized study reported that prednisolone and azathioprine preserved kidney function and improved histologic appearances, but it relied on historical controls.⁶¹

Slowly Progressive Chronic Kidney Disease

Although the kidney histology and clinical course of slowly progressive IgAV and IgAN may be indistinguishable, patients with IgAV have not been included in studies of slowly progressive IgAN. Therefore, there is no evidence that fish oil is beneficial in IgAV. Tight BP control with ACE inhibitors or angiotensin receptor blockers (ARBs) is recommended for proteinuric IgAV as for IgAN. The 2021 KDIGO guidelines recommend a similar indication for corticosteroids in slowly progressive IgAV as in patients with intermediate-risk IgAN.^{30a}

Transplant Recurrence

No treatment is known to reduce the risk for recurrence. There is some evidence that recurrence is more common and more likely to lead to graft loss in children receiving kidneys from living rather than deceased donors, although this is not confirmed in adults.^{62,63} If crescentic IgAV recurs, intensive immunosuppression may be justified as for primary disease. This, however, has not been thoroughly evaluated.

SELF-ASSESSMENT QUESTIONS

- Based on the 2021 KDIGO guidelines, which treatment is recommended for all patients at risk for progressive loss of *GFR*?
 - Fish oil
 - Mycophenolate mofetil
 - Prednisolone
 - Renin angiotensin aldosterone system inhibitor
 - Tonsillectomy
- The Oxford (MEST-C) classification includes pathologic features predictive of outcome. Which of the following is *not* one of the predictive features?
 - Endocapillary hypercellularity
 - Capillary vascular changes (nephrosclerosis)
 - Mesangial hypercellularity
 - Segmental glomerulosclerosis
 - Tubulointerstitial damage and fibrosis
- Which of the following statements for IgA nephropathy is *true*?
 - IgAN is more common in Europe than any other part of the world.
 - Measurement of abnormally glycosylated serum IgA is a diagnostic test for IgAN.
 - The renal pathologic features can be identical to those in IgA vasculitis.
 - Acute kidney injury with macrohematuria in IgA nephropathy is almost always caused by crescentic IgAN.
 - The risk for recurrence after kidney transplantation is reduced when MMF is part of the immunosuppressive regimen.
- In IgA vasculitis, which of the following statements is *true*?
 - IgA vasculitis never occurs after age 40 years.
 - There is altered glycosylation of serum IgA1.
 - Delay in transplantation reduces the risk for recurrence.
 - Cyclophosphamide is of proven benefit for slowly progressive IgA vasculitis.
 - Corticosteroids given at the onset of the rash reduce the risk for subsequent nephritis.

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